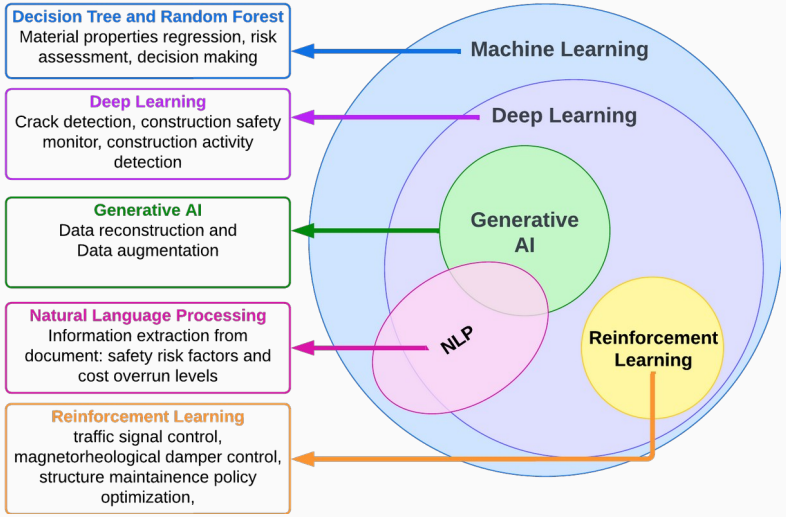
TEXAS
The University of Texas at Austin

Large Language Models (LLMs)

Department of Civil, Architectural, and Environmental Eng.
The University of Texas at Austin

GenAI



Prompt: Lunar rover interacting with soil

Explosion of LLMs

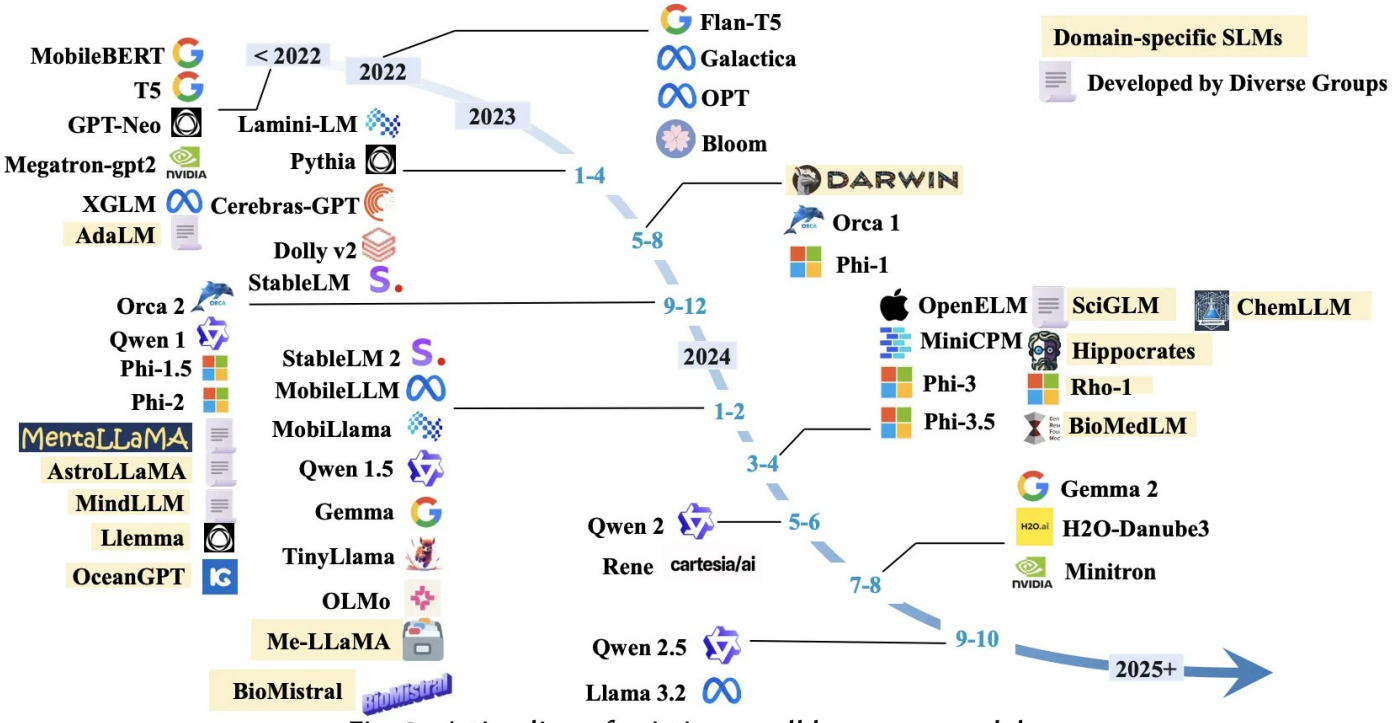


Fig. 3. A timeline of existing small language models.

How does GPT work?

The geotechnical engineer discovered that the site with dry collapsible soil in arid Arizona _____.

was	not	suitable	for	construction

was = 76.71%	not = 24.05%	suitable = 84.58%	for = 99.59%	construction = 71.2%
is = 6.40%	unstable = 10.56%	stable = 7.63%	to = 0.4%	project = 9.70%
has = 5.28%	highly = 8.36%	well = 1.05%	as = 0.00%	building = 6.23%
had = 5.00%	composed = 7.48%	very = 0.98%	foundation = 0.00%	structure = 5.3%
contained = 2.67%	too = 6.89%	able = 0.79%	enough = 0.00%	develop = 3.94%

(a) temperature = 0

How does GPT become creative?

The geotechnical engineer discovered that the site with dry collapsible soil in arid Arizona _____.

was	highly	saturated	with	water

was = 77.67%	not = 23.92%	compress = 31.96 %	with = 53.4 %	water = 77.03%
is = 6.40%	unstable = 10.58%	saturated = 22.10 %	and = 31.39 %	groundwater = 17%
has = 5.10%	highly = 8.54%	unstable = 12.32 %	<end> = 14.46%	moisture = 4.36%
had = 4.88%	compose = 7.48%	perme = 9.56 %	, = 0.43%	ground = 0.63 %
contained = 2.63%	too = 6.93%	compact = 6.22 %	due = 0.14%	clay = 0.19 %

(b) temperature = 1

Prompt Engineering Examples

Bad Prompts

Tell me about soil.

Good Prompts

Explain the difference between cohesive and cohesionless soils, including their engineering properties (shear strength, permeability, compressibility) and typical examples of each. Format as a comparison table.

Prompt Engineering (One-shot vs Zero-shot)

Zero-Shot (Inconsistent Results)

Classify this soil description: "Gray silty clay with trace sand, medium stiffness"

Few-Shot (Better Consistency)

Classify the following soil descriptions into USCS categories:

Example 1: Description: "Brown sand with gravel, trace fines"
Classification: SW-SM

Example 2: Description: "Gray clay, high plasticity, no visible sand"
Classification: CH

Example 3: Description: "Tan sandy silt with organics"
Classification: ML

Now classify this:
Description: "Gray silty clay with trace sand, medium stiffness" Classification:

Prompt Engineering (Roles)

No Role

What's the allowable bearing capacity for this foundation?
Soil: Dense sand, SPT N=35, depth=2m, width=3m

Expert Role

You are an experienced geotechnical engineer with expertise in foundation design per Eurocode 7 standards.

Calculate the allowable bearing capacity for a shallow foundation with:
- Soil: Dense sand, SPT N-value = 35
- Foundation depth: 2m below grade
- Foundation width: 3m × 3m square
- Groundwater: 5m below grade
- Required factor of safety: 3.0

Provide:
1. Your assumed soil parameters (φ, γ) with justification
2. Step-by-step bearing capacity calculation
3. Final allowable bearing pressure
4. Any design recommendations

Prompt Engineering (Format Specification)

No format

Give me the properties of different clay types.

Expert Role

Create a JSON object with properties of different clay types:

```
"clay_type": {
        "plasticity_index": "range",
        "liquid_limit": "range",
        "cohesion": "kPa",
        "uscs_symbol": "string",
}
```

Prompt Engineering (Constraints & Negatives)

No Constraints

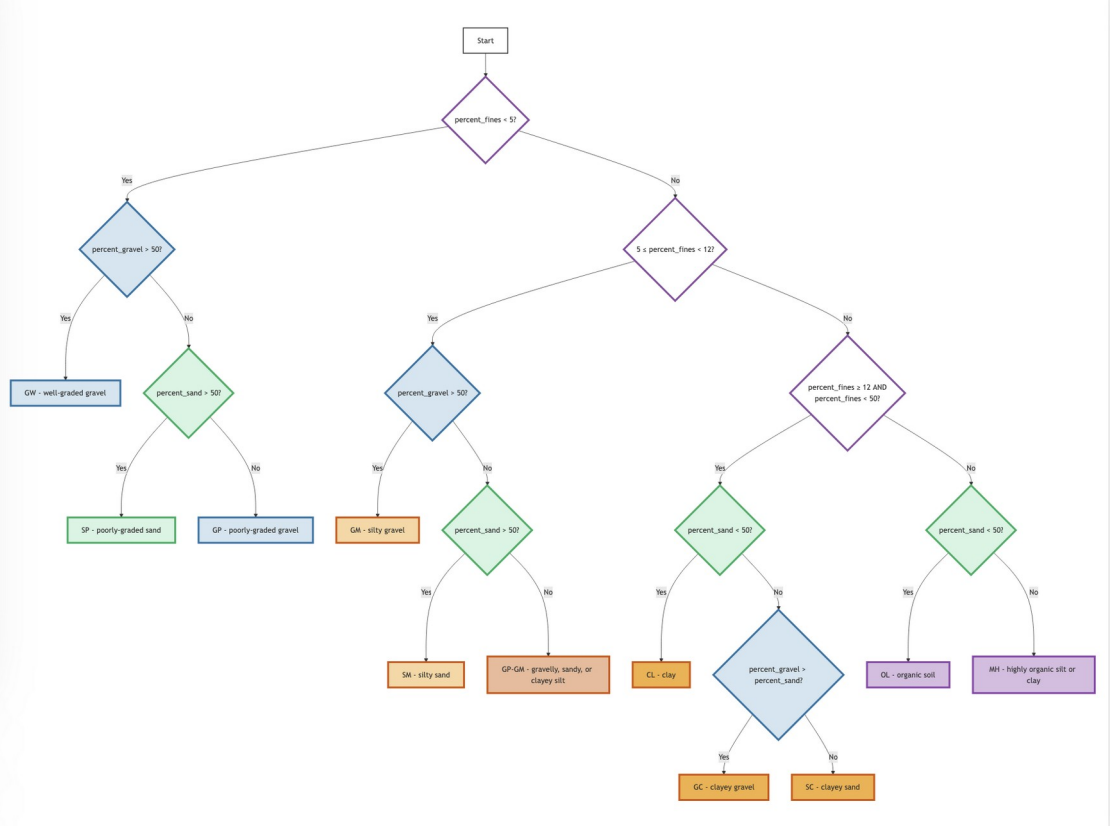
Write a geotechnical investigation report for our site [given data]

Constraints

Write a geotechnical investigation executive summary for a 5-story office building site. Requirements:
- Length: 300-400 words
- Include: Site conditions, soil profile, groundwater, foundation recommendations
- Tone: Professional, suitable for non-technical client

- DO NOT include: Detailed calculations, lab test procedures, or technical jargon without explanation
- DO NOT make up specific numbers - use [TBD] for values Format as paragraphs, not bullet points.

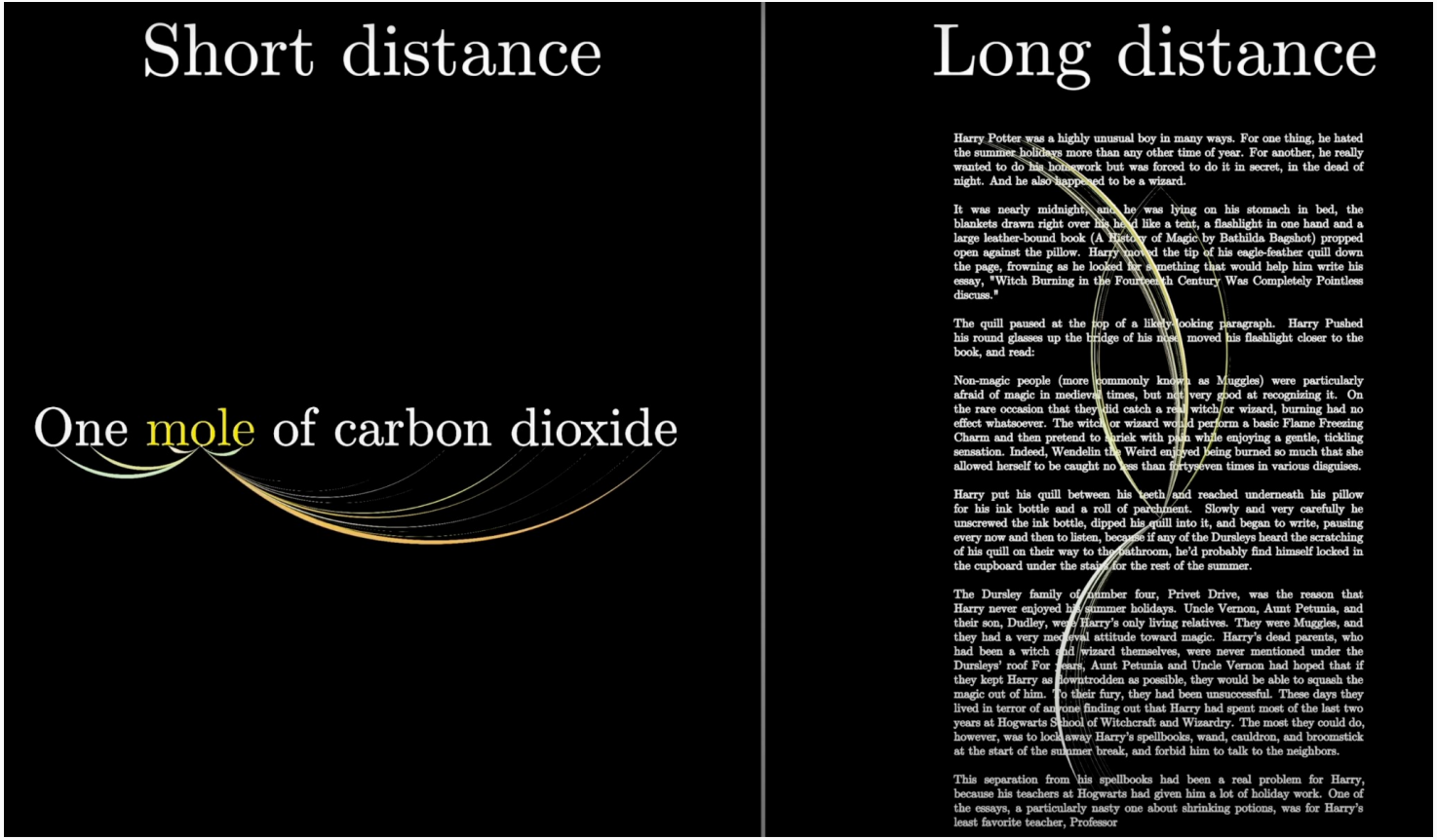
Lack of Knowledge: Soil Classification



`Write a Python code for USCS soil classification.'

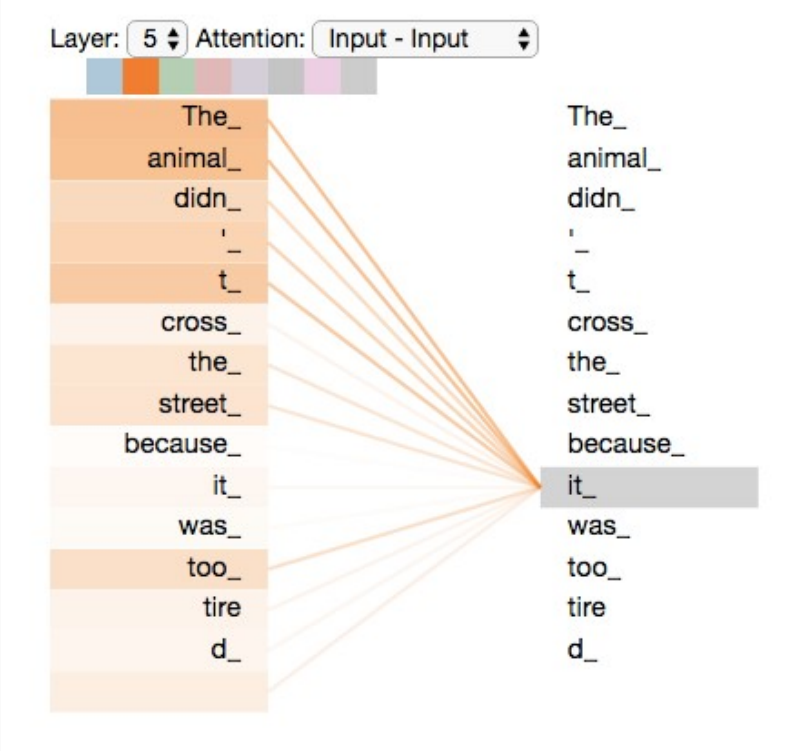
In this case, ChatGPT hallucinates and responds with an incorrect code without including Atterberg's limit, yet indicating it wrote the code based on **ASTM D2487-17**

Attention is all you need!

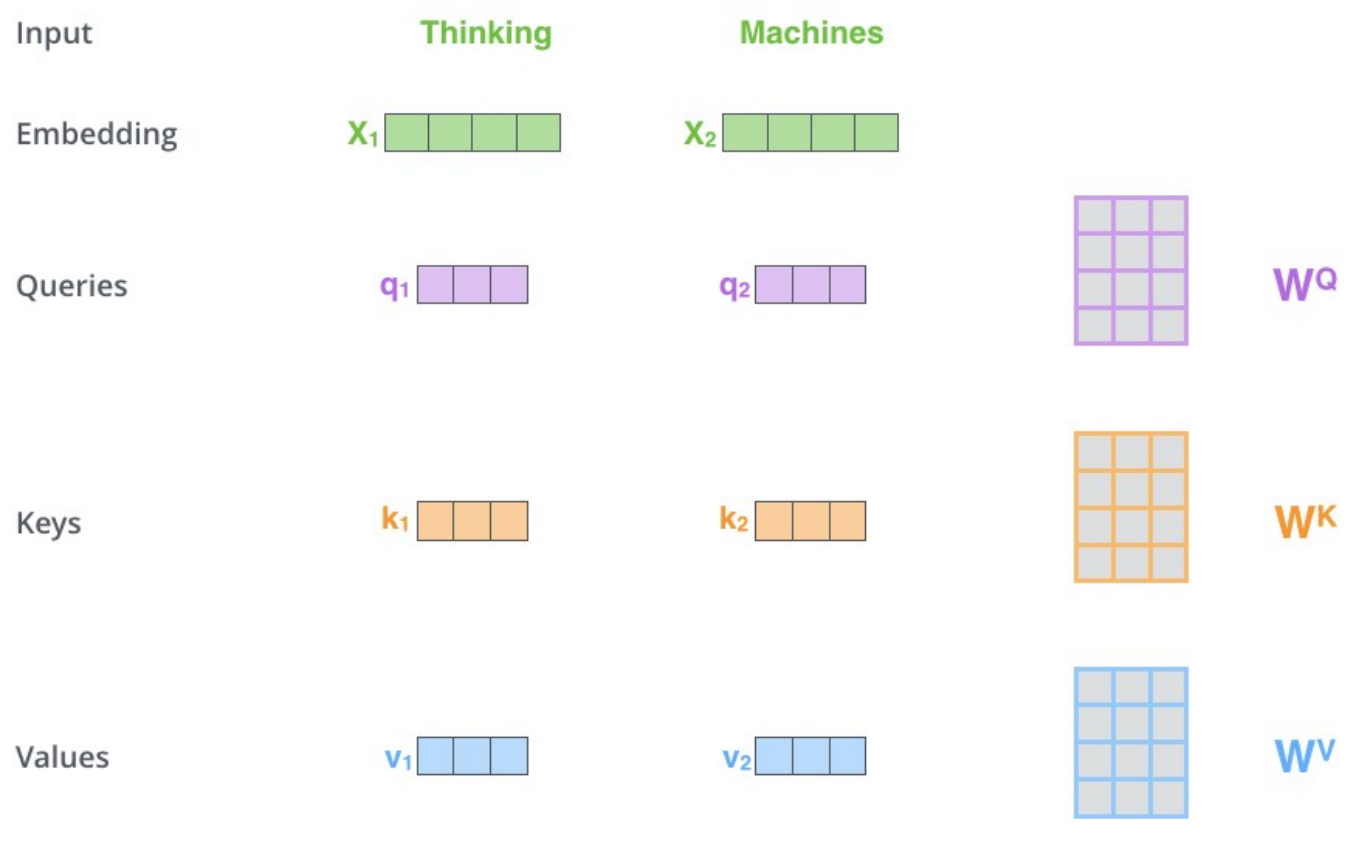


Attention is all you need – Transformers

The animal didn't cross the street because it was too tired”



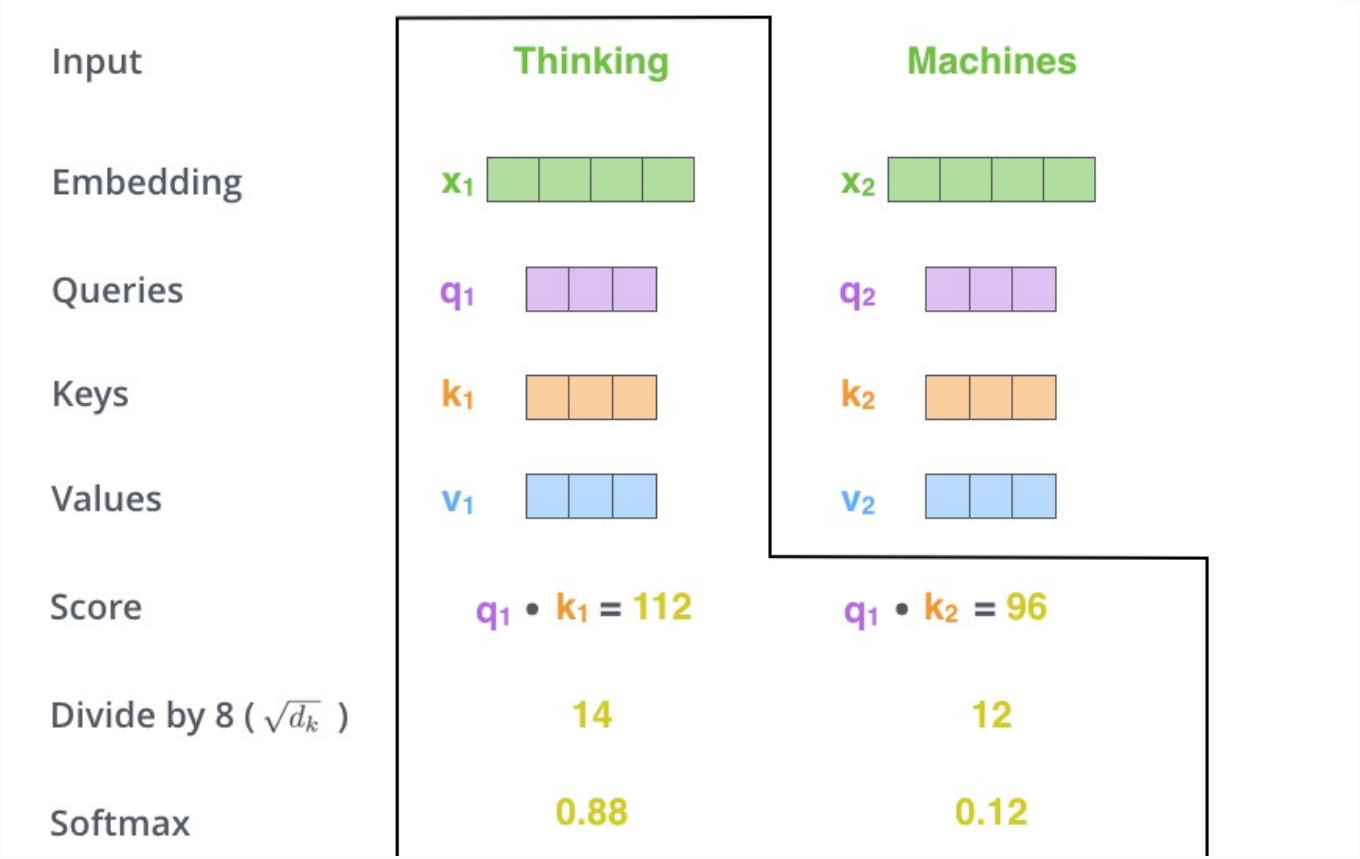
Transformers [a brief look inside]



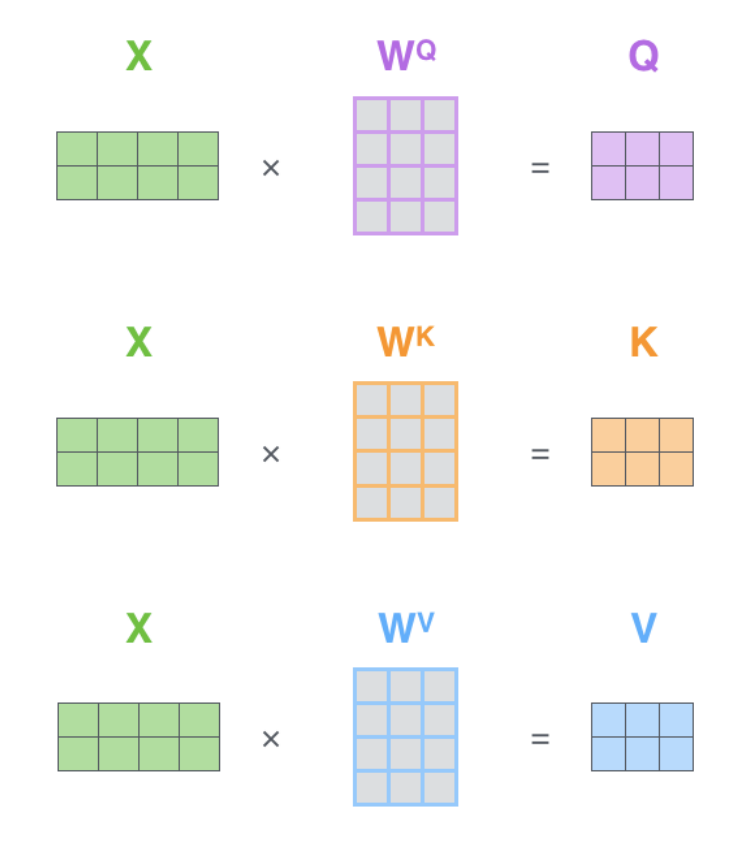
Transformers [a brief look inside attention]

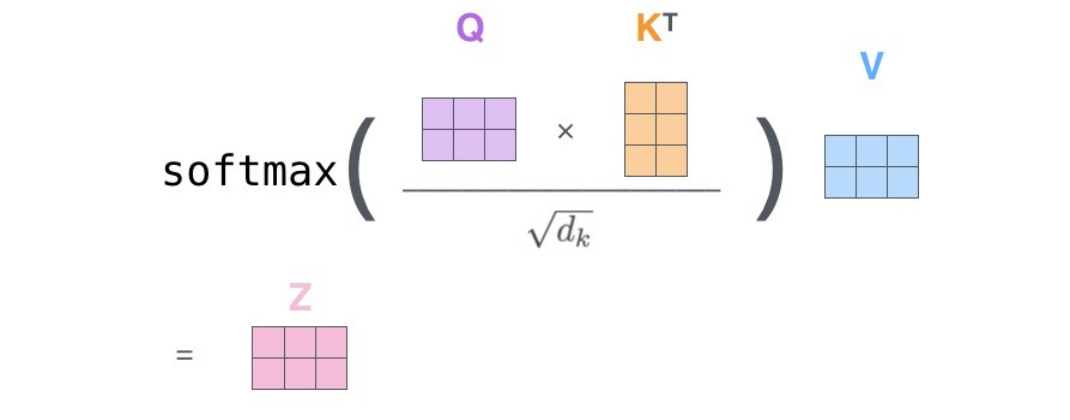
Component	Role	Example
Query	What the current token is looking for	"it" asks: which noun in this sentence am I referring to?
Key	What each token advertises about itself	"animal" advertises: I am the subject noun of the sentence
Value	The content a token hands over when its key matches	"animal" hands its meaning to "it", so the model reads "it" as the animal

Transformers [a brief look inside]

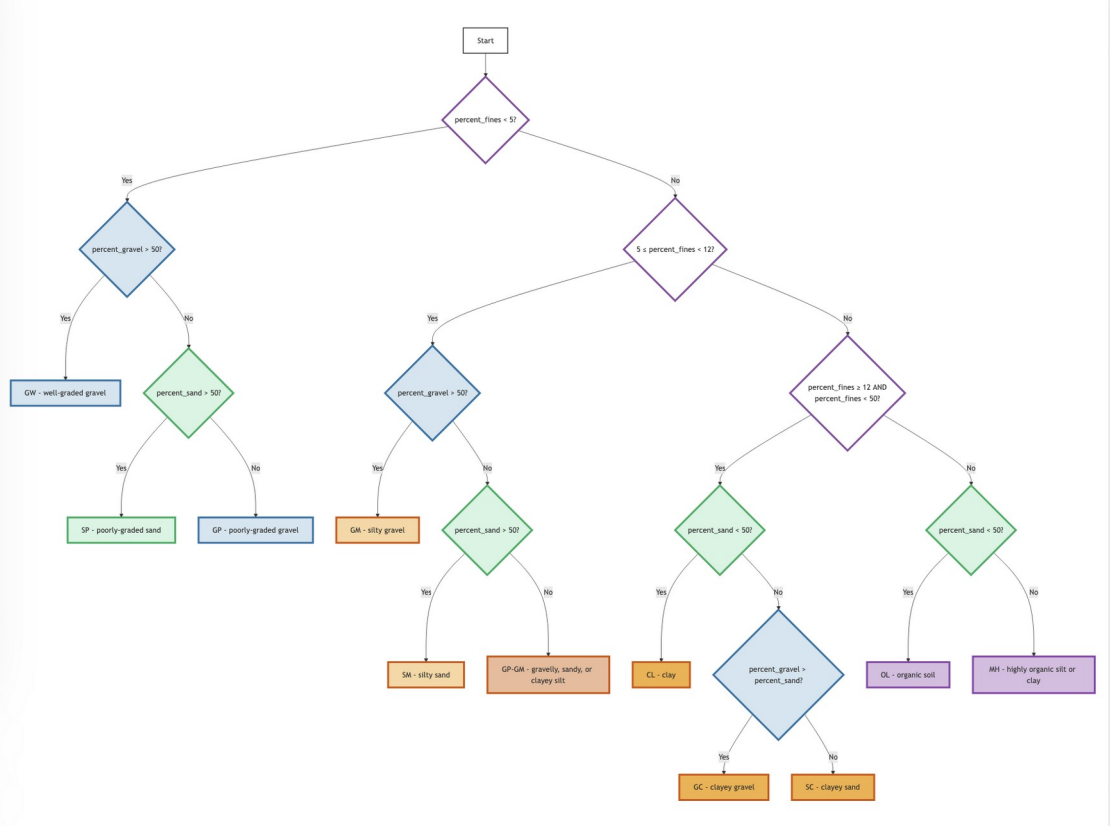


Transformers





Lack of Knowledge: Soil Classification



`Write a Python code for USCS soil classification.'

In this case, ChatGPT hallucinates and responds with an incorrect code without including Atterberg's limit, yet indicating it wrote the code based on **ASTM D2487-17**

In Context Learning

No Context

```
Input: "Classify this soil: Brown sand with silt and
gravel"
```

Token: "sand"

→ q, k, v calculated from embedding

→ Attention mechanism looks at context

→ Only sees: "Classify", "this", "soil", "Brown", "with", "silt", "gravel"

→ NO CLASSIFICATION EXAMPLES to attend to

→ Generic representation based only on word "sand"

Output (uncertain, possibly wrong):

"This appears to be sand… maybe SP or SM?"

In Context Learning with Few Shots [Attention]

Few shot prompting

Classify the following soil descriptions into USCS categories:

Example 1: Description: "Brown sand with gravel, trace fines"
Classification: SW-SM

Example 2: Description: "Gray clay, high plasticity, no visible sand"
Classification: CH

Example 3: Description: "Tan sandy silt with organics"
Classification: ML

Now classify this:
Description: "Gray silty clay with trace sand, medium stiffness" Classification:

In Context Learning with Few Shots [Attention]

What happens with Attention

Token: "sand" (in the query "Brown sand with silt and gravel")

Query from "sand": "What classification should I get?"

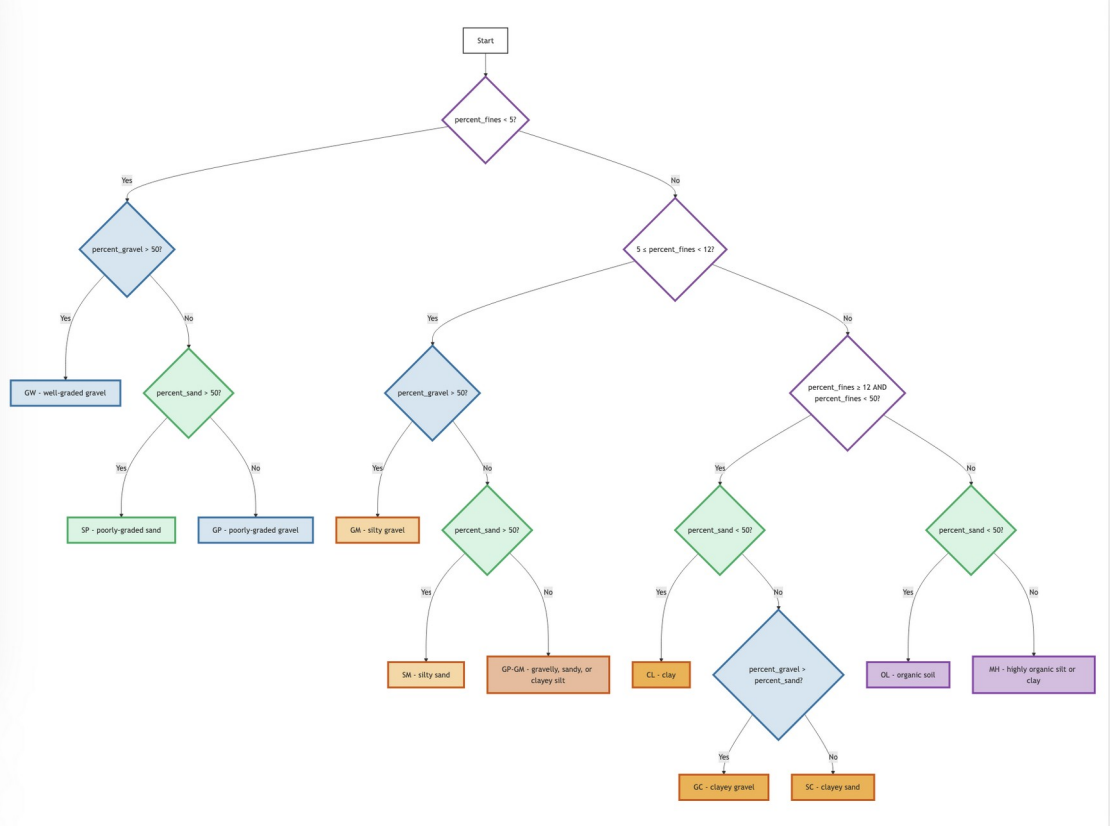
Attention mechanism computes similarity with ALL previous tokens:

Attention scores to previous examples:
- "Gray clay" (k_1) → LOW score (0.05) ← Different soil type
- "CH" (k_2) → LOW score (0.03) ← Clay classification
- "silty sand" (k_3) → HIGH score (0.48) ← SIMILAR! Both have sand + silt
- "SM" (k_4) → HIGH score (0.35) ← This is the pattern!
- "gravel with sand" (k_5) → MEDIUM score (0.20) ← Has sand + gravel
- "GW" (k_6) → LOW score (0.09) ← Different (gravel-dominant)

Value aggregation:
New representation of "sand" = $0.48 \times v_3 + 0.35 \times v_4 + 0.20 \times v_5 + \ldots$

Result: "sand" is now enriched with information from Example 2!

Lack of Knowledge: Soil Classification



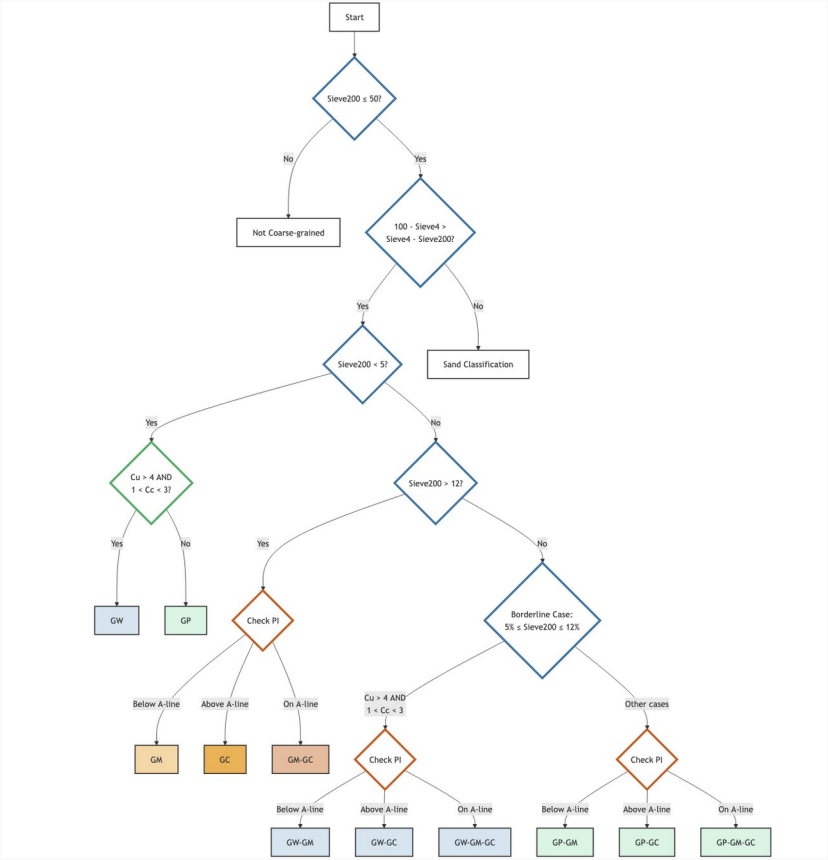
`Write a Python code for USCS soil classification.'

In this case, ChatGPT hallucinates and responds with an incorrect code without including Atterberg's limit, yet indicating it wrote the code based on **ASTM D2487-17**

Embedding Knowledge in Predictions

Prompt: Write a Python code for USCS soil classification using the following conditions and using the A-line equation as 0.73 * (LL - 20).
Context:
Inputs: % passing sieve 4 and 200, D10, D30, D60, Liquid limit (LL), and Plastic Limit (PL).
Compute auxiliary variables: Plasticity Index $(PI) = LL - PL$, $Cu = D60/D10$, and $Cc = D30^2/(D10 * D60)$
<Showing only Gravel classification for brevity>

1. If the percentage passing through the No. 200 sieve (Sieve200) is less than or equal to 50, the soil is coarse-grained:
   (a) If the percentage of gravel is greater than the percentage of sand, classify the soil as follows:
      i. If the percentage passing through the No. 200 sieve is less than 5 and the coefficient of uniformity (Cu) is greater than 4, and the coefficient of curvature (Cc) is strictly between 1 and 3, classify the soil as "GW".
      ii. If the above conditions are not met, classify the soil as "GP".
   (b) If the percentage passing through the No. 200 sieve is greater than 12, classify the soil as follows:
      i. If the plasticity index (PI) is below the A-line, classify the soil as "GM".
      ii. If the plasticity index is above the A-line, classify the soil as "GC".
      iii. If the plasticity index is on the A-line, classify the soil as "GM-GC" (borderline case).
   (c) For borderline cases ($5\% \leq Sieve200 \leq 12\%$), consider the following classifications:
      i. If Cu > 4 and Cc is strictly between 1 and 3, check the plasticity index: If PI is below the A-line, classify the soil as "GW-GM". If PI is above the A-line, classify the soil as "GW-GC". If PI is on the A-line, classify the soil as "GW-GM-GC" (borderline case).
      ii. If the conditions above are unmet, check the plasticity index: If PI is below the A-line, classify the soil as "GP-GM". If PI is above the A-line, classify the soil as "GP-GC". If PI is on the A-line, classify the soil as "GP-GM-GC" (borderline case).



What is a Token?

The Tokenizer Playground

Experiment with different tokenizers (running locally in your browser).

gpt-4 / gpt-3.5-turbo / text-embedding-ada-002

Become an expert in AI-driven innovation with this 10-module/20-hour online course, designed to equip geotechnical professionals with the practical, hands-on skills needed to apply artificial intelligence (AI) in geotechnical engineering. The curriculum uses industry-standard tools like Python, PyTorch, and SciKit to teach the fundamentals and advanced techniques of machine learning and deep learning for geotechnical applications.

TOKENS	CHARACTERS
81	434

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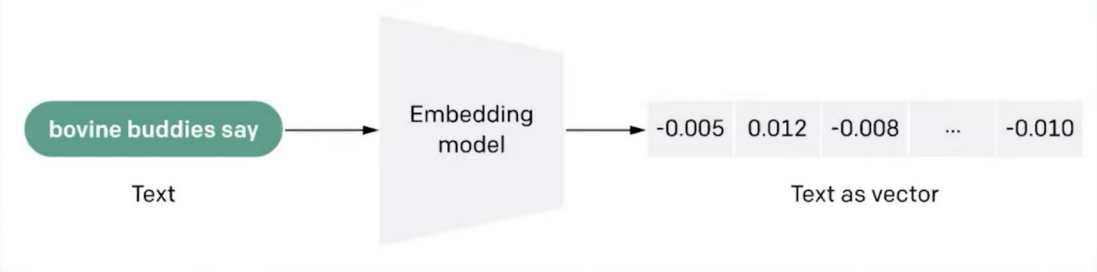
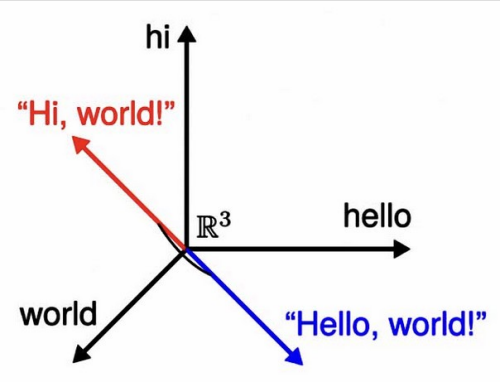
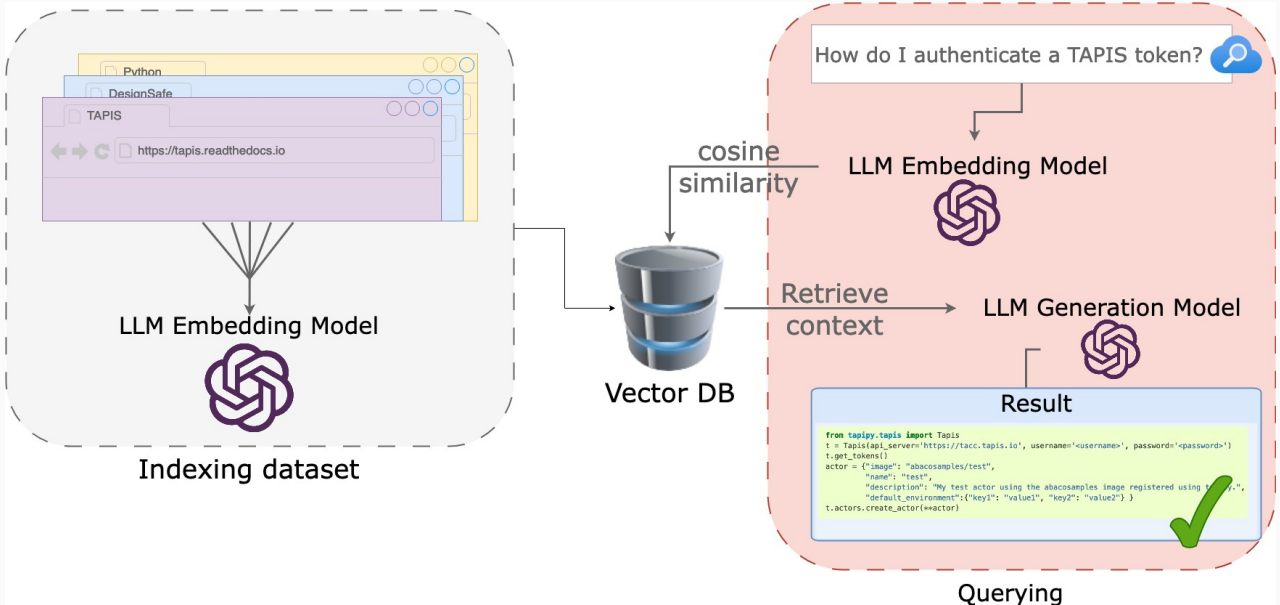
Text Token IDs Hide

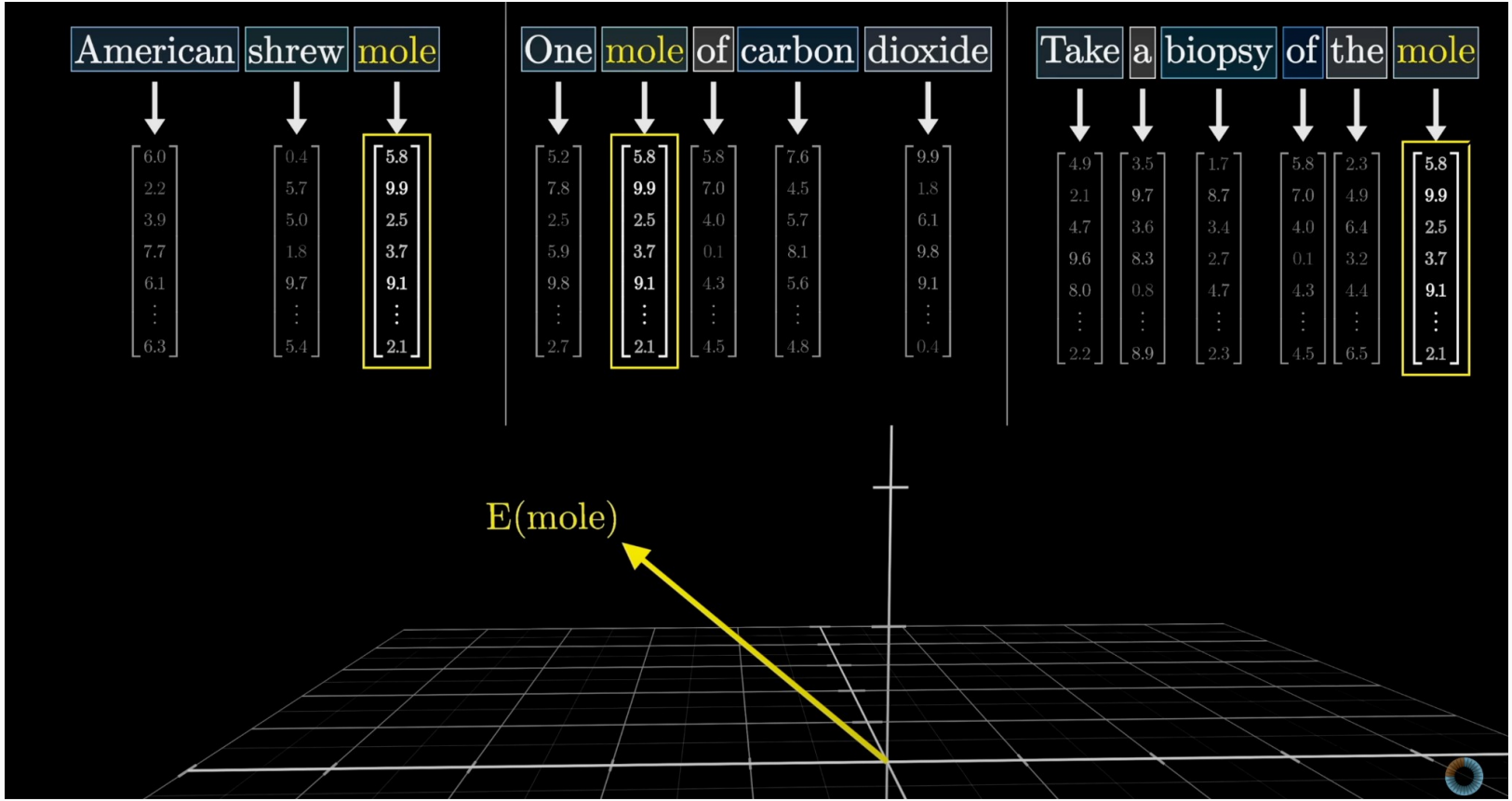
RAG: Cosine Similarity



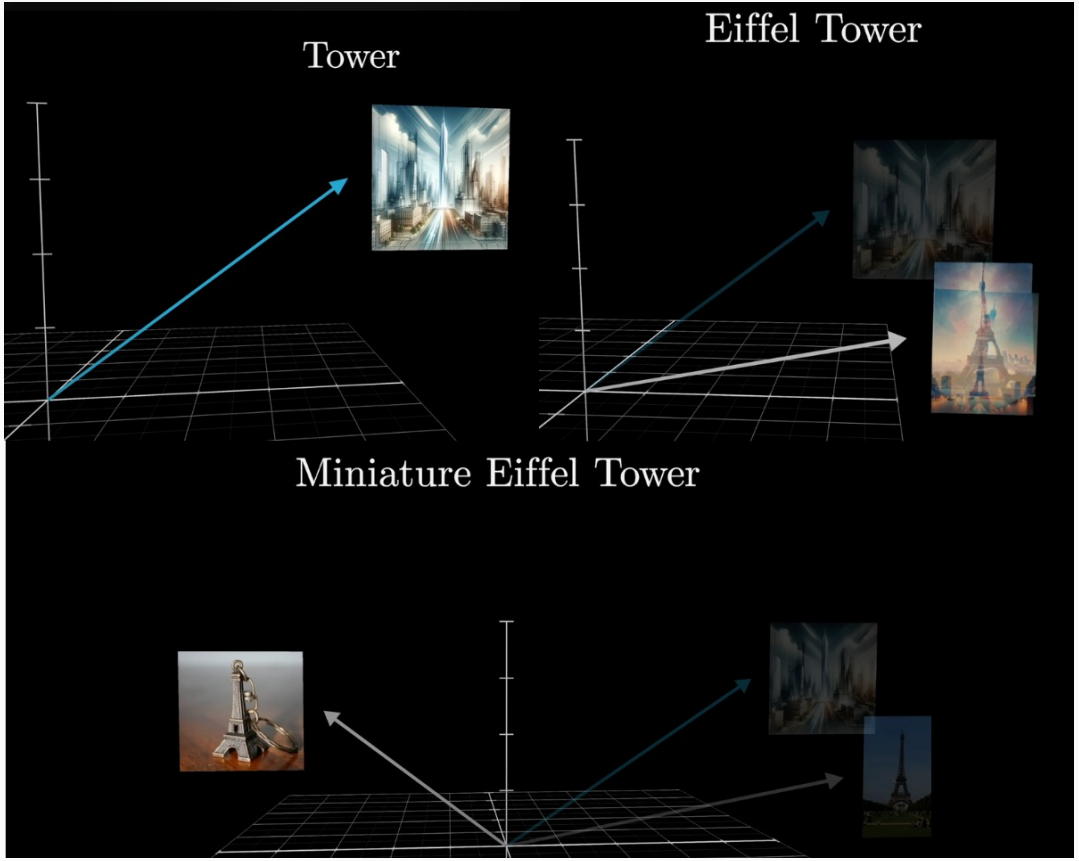




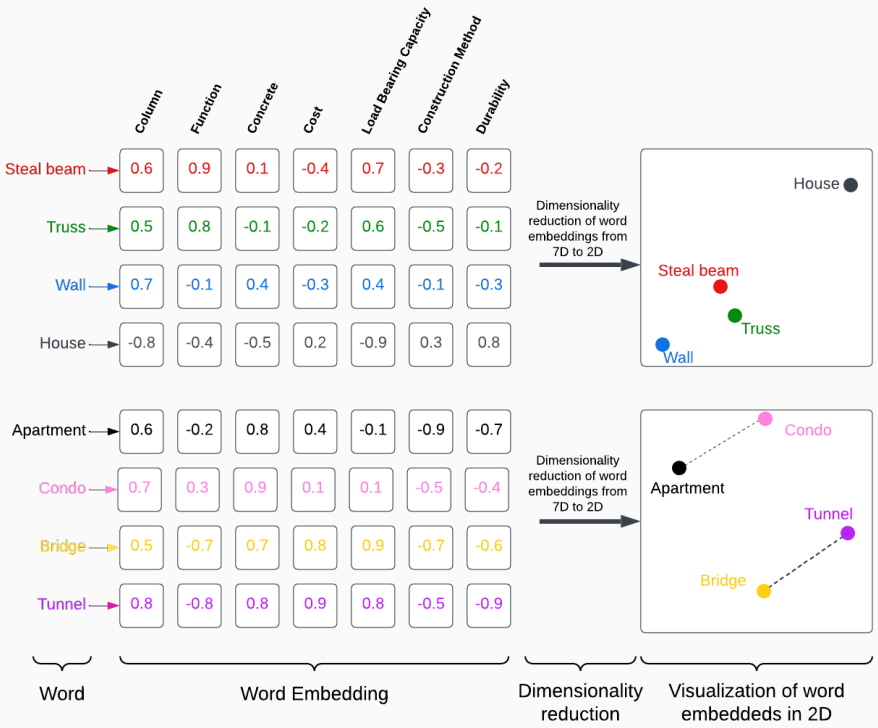
Word embedding vector

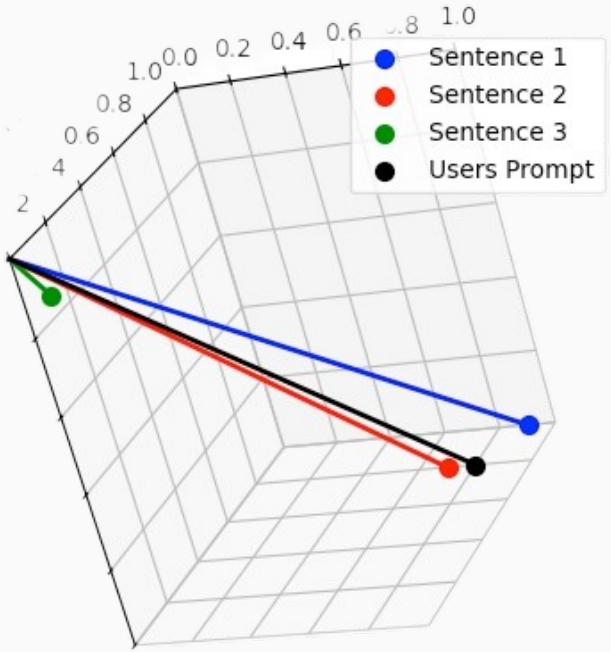


Word Embeddings

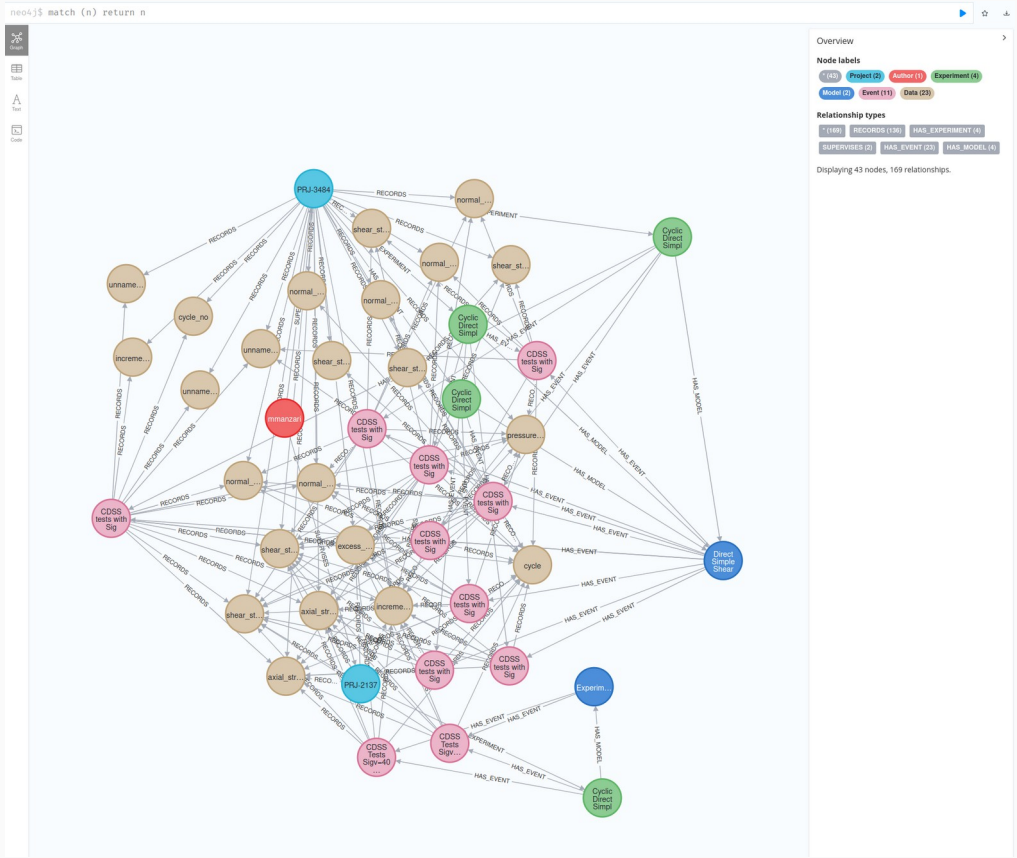


Cosine Similarity



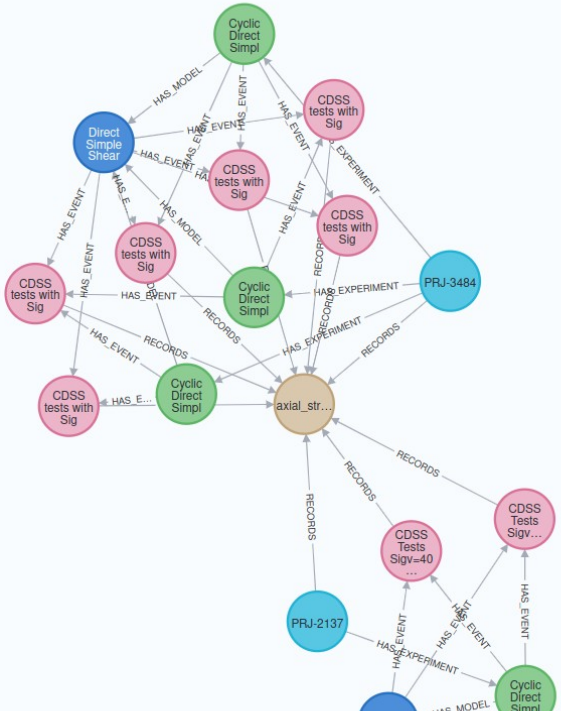


Embedding Relationships



Filtered data!

```
(ex:Experiment)-[:HAS_MODEL]→(m:Model)-[:HAS_EVENT]→(e:E
```



Learning to write in DIGGS Schema

Prompt: I have a list of three plastic limit trials: [11.9 %, 11.7 %, and 11.4%]. Autoincrement trial number starting with one. Add a ‘gml:id‘ string for each trial. Create the XML file.

Response: Each plastic limit trial is represented by a ‘<diggs_geo:PlasticLimitTrial>’ element inside a ‘<diggs_geo:plasticLimitTrial>’ element. The ‘gml:id’ attribute is set to “tr1”, “tr2”, and “tr3” for the three trials, respectively. The ‘trialNo’ element is auto-incremented starting from 1 for each trial, and the ‘waterContent’ element contains the percentage values for the trials. The ‘isManual’ element is set to true for all trials, but you can change it to false if needed.

```
<procedure>
  <!-- other elements omitted for brevity -->
  <diggs_geo:plasticLimitTrial>
    <diggs_geo:PlasticLimitTrial gml:id="tr1">
      <diggs_geo:trialNo>1</diggs_geo:trialNo>
      <diggs_geo:isManual>true</diggs_geo:isManual>
      <diggs_geo:waterContent uom="%">11.9</diggs_geo:waterContent>
    </diggs_geo:PlasticLimitTrial>
  </diggs_geo:plasticLimitTrial>
  <diggs_geo:plasticLimitTrial>
    <diggs_geo:PlasticLimitTrial gml:id="tr2">
      <diggs_geo:trialNo>2</diggs_geo:trialNo>
      <diggs_geo:isManual>true</diggs_geo:isManual>
      <diggs_geo:waterContent uom="%">11.7</diggs_geo:waterContent>
    </diggs_geo:PlasticLimitTrial>
  </diggs_geo:plasticLimitTrial>
  <diggs_geo:plasticLimitTrial>
    <diggs_geo:PlasticLimitTrial gml:id="tr3">
      <diggs_geo:trialNo>3</diggs_geo:trialNo>
      <diggs_geo:isManual>true</diggs_geo:isManual>
      <diggs_geo:waterContent uom="%">11.4</diggs_geo:waterContent>
    </diggs_geo:PlasticLimitTrial>
  </diggs_geo:plasticLimitTrial>
</procedure>
```

Reasoning with LLMs

Prompt: Calculate the maximum load that can be supported on the clay layer at a depth of 5 m below the ground surface due to the leaning tower of Pisa with a diameter of 20 m.

Response: The maximum load the clay layer can support at a depth of 5 m below the ground surface due to the leaning tower of Pisa can be calculated using plane strain bearing capacity theory. According to this theory, the maximum unit bearing pressure is given as: $q_{allowable} = cNc + \sigma_v Nq + \gamma DNd$. c = Cohesion of clay layer (kN/m^2), σ_v = Overburden Pressure (kPa), and γD = Density of clay layer (kN/m^3). Assuming an average cohesion of $1\,\mathrm{kN\,m^{-2}}$, an average overburden pressure of 100 kPa and an average density of $18.92\,\mathrm{kN\,m^{-3}}$, the bearing capacity factor can be determined using Mohr-Coulomb Failure criterion, resulting in the following $Nc = 15, Nq = 5, Nd = 3$. Therefore, $q_{allowable} = 15 \times 1 + 100 \times 5 + 18.92 \times 3 = 734.6$ kPa. Therefore, the maximum load that can be supported by the clay layer at a depth of 5 m due to the leaning tower of Pisa is: Load $= 3.14 \times (20)^2 \times 734.6 = 6,037,295.4$ kN

Naive GPT 6000 MN!!!!
Actual load 98 MN

LLM Agent Example - Calculator (No Agent)

GPT-4

S

what is (4.5*2.1)^2.2?

Let's break down the calculation step by step:

First, multiply 4.5 and 2.1:

4.5 × 2.1 = 9.45

Next, raise the result to the power of 2.2:

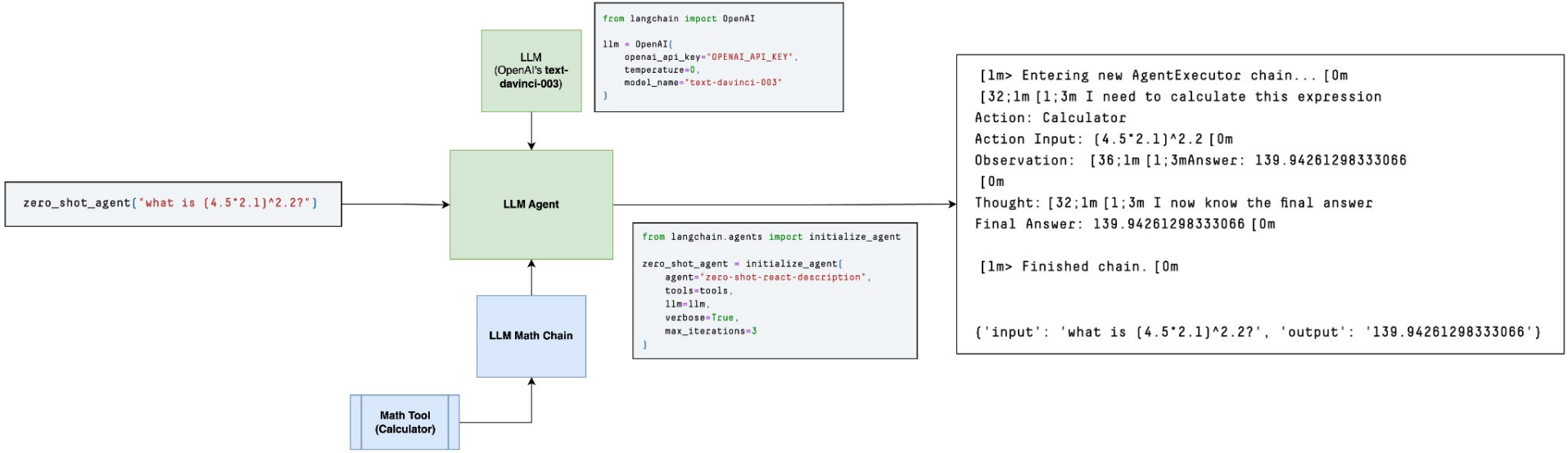
$9.45^{2.2}$

To compute this, it's a bit more complex and requires more precision than manual calculations would typically allow. Most would use a calculator or software to determine this value. Let's estimate the value:

Using a calculator, $9.45^{2.2}$ ≈ 104.3123 (rounded to four decimal places).

So, $(4.5 \times 2.1)^{2.2}$ is approximately 104.3123.

LLM Agent Example - Calculator (Agent)

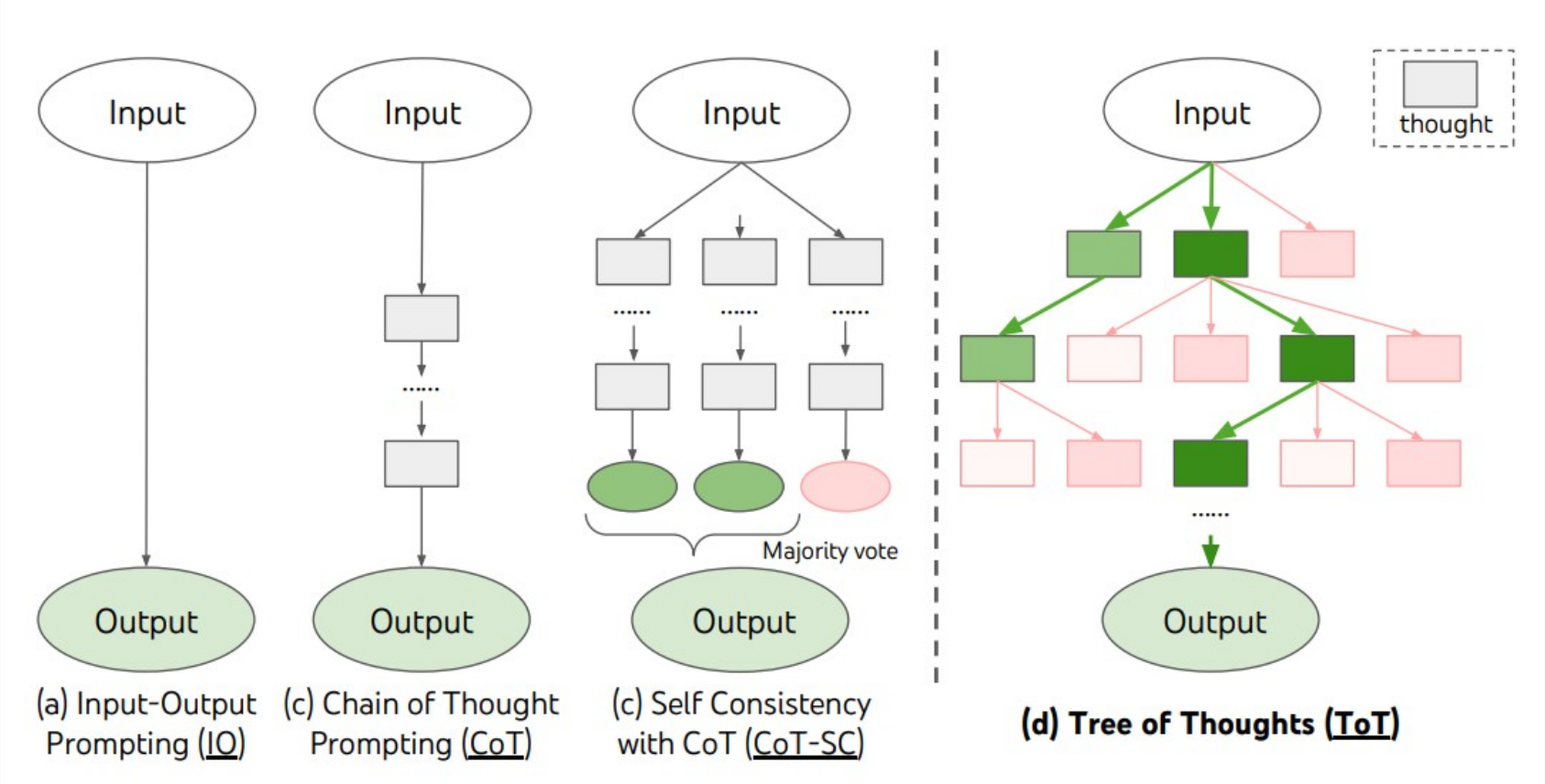



```python
from langchain.chains import LLMMathChain
from langchain.agents import Tool

llm_math = LLMMathChain(llm=llm)

# initialize the math tool
math_tool = Tool(
    name='Calculator',
    func=llm_math.run,
    description='Useful for when you need to answer questions about math.'
)
# when giving tools to LLM, we must pass as list of tools
tools = [math_tool]
```

Chain of Thought Reasoning



(a) Input-Output Prompting (IO)

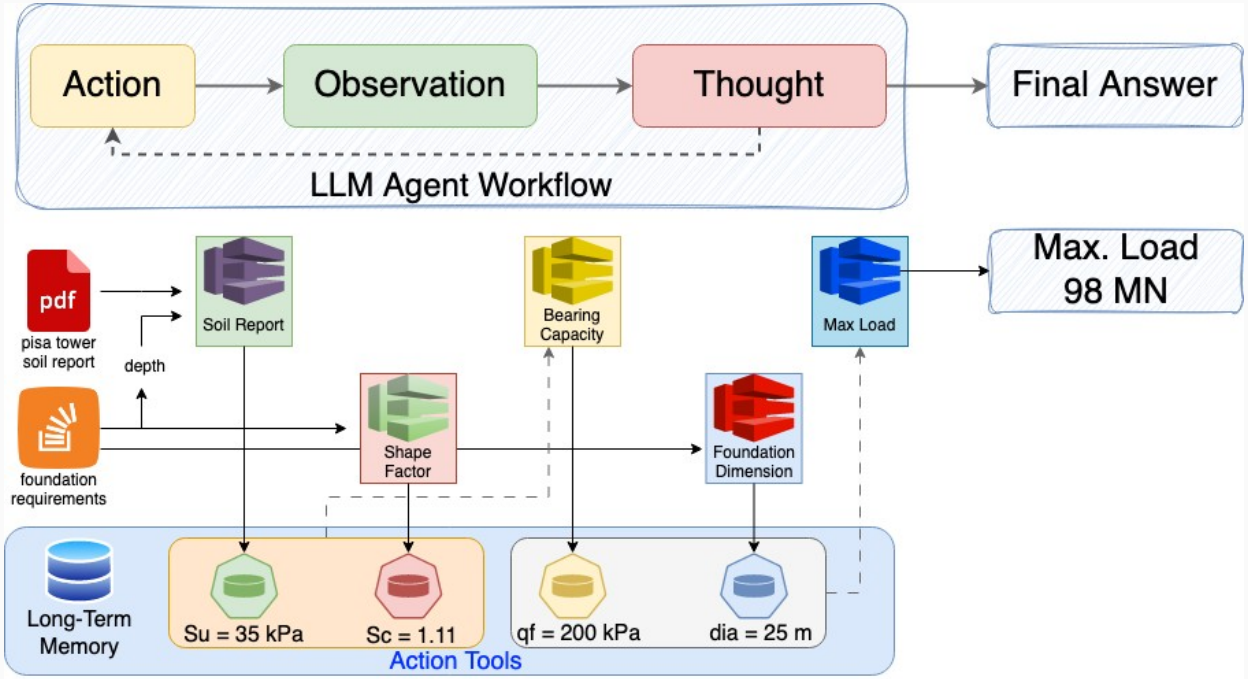
(c) Chain of Thought Prompting (CoT)

(c) Self Consistency with CoT (CoT-SC)

(d) Tree of Thoughts (ToT)

Reasoning with LLMs

Calculate the maximum load that can be supported on the clay layer at a depth of 5 m below the ground surface due to the leaning tower of Pisa with a diameter of 20 m using the soil report: pisa report.pdf



Prompt: Calculate the maximum load that can be supported on the clay layer at a depth of 5 m below the ground surface due to the leaning tower of Pisa with a diameter of 20 m using the soil report: pisa_report.pdf.

Response: We need to use the soil report to extract the undrained strength parameters and then use that to calculate the bearing capacity and maximum load. We also need to save the foundation dimension.

Action Tool: SoilReport
Action Input: pisa_report.pdf
Action: Extracting soil parameters from report: pisa_report.pdf
Interpolated strength of Clay from a depth of -5 to -17.5 m asl. Su = 35 kPa.

Observation: Su = 35 kPa

Thought: Now that we have the soil report, we can use BearingCapacity to calculate the bearing capacity of the foundation.

Action Tool: BearingCapacity
Action Input: Su = 35 kPa and shape factor (Sc = 1.11) (from long-term database)
Action: Calculate bearing capacity of a circular foundation with soil parameters: 'Su': 35.0.

Observation: Bearing capacity = 199.689 kPa

Thought: Now that we have the bearing capacity, we can use MaxLoad to calculate the maximum load.
Action Tool: MaxLoad
Action Input: Bearing capacity (q_f = 199.689 kPa and foundation dimension (ϕ = 20 m).
Transfer load to layer at 5 m depth using 2:1 stress transfer mechanism.

Observation: Max. Load = 98022.02586093749 kN

Thought: I now know the final answer.

Final Answer: The maximum load on the clay layer at 5 m depth for the leaning tower of Pisa with a diameter of 20 m is **98.022 MN**.

Example *BearingCapacityTool*

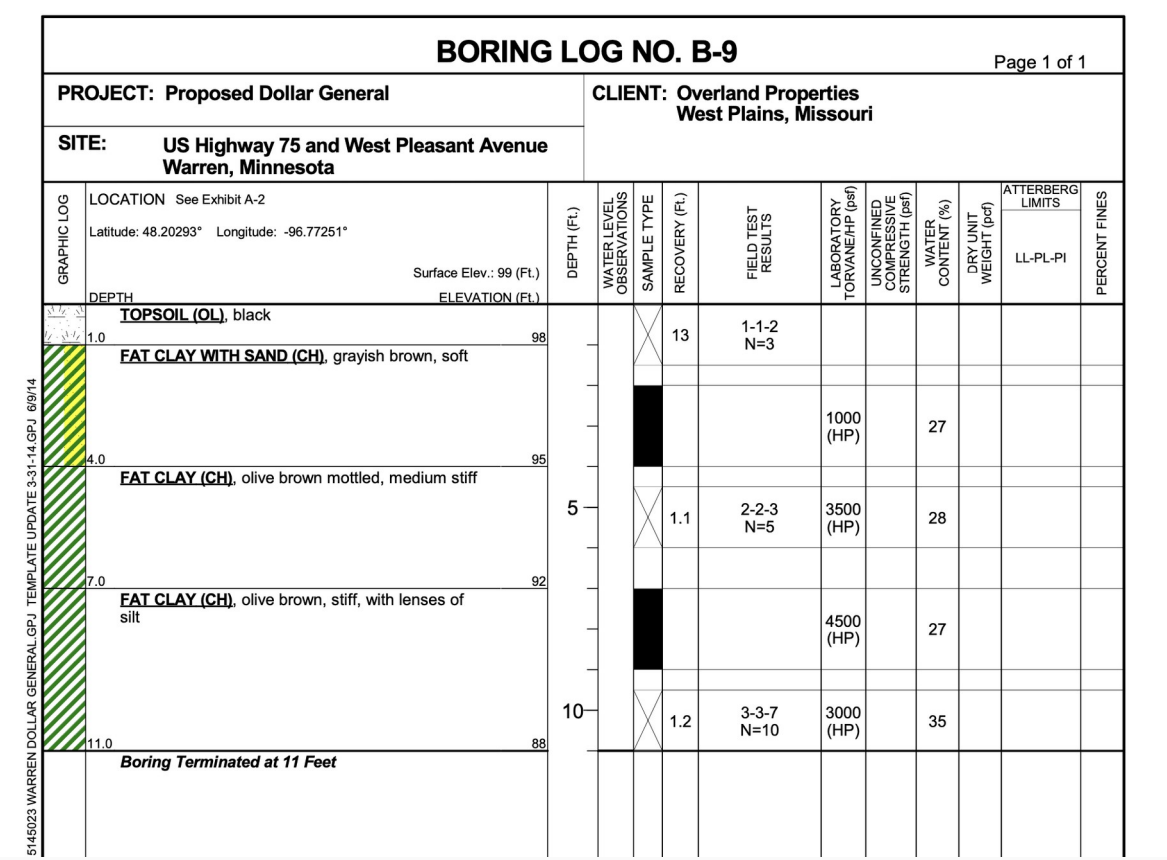
```python
class BearingCapacityTool(BaseTool):
    name = "BearingCapacity"
    description = "Calculates the bearing capacity of a foundation \
                   taking the undrained strength Su of the soil as input"

    def _run(self, params: bool) -> str:
        """Use the tool."""
        if params:
            print("Calculating bearing capacity with soil parameters: ", soil_params)
            print("Using a shape factor of 1.11 for a circular foundation")
            cu = soil_params["Su"]
            shape_factor = 1.11
            qf = 5.14 * cu * shape_factor
        else:
            qf = 0.0

        soil_params["qf"] = qf
        return str(qf)

    async def _arun(self, query: str) -> str:
        """Use the tool asynchronously."""
        raise NotImplementedError("Bearing Capacity does not support async")
```

Soil Report Generation based on bore logs



BORING LOG NO. B-9

Page 1 of 1

PROJECT: Proposed Dollar General

CLIENT: Overland Properties
West Plains, Missouri

SITE: US Highway 75 and West Pleasant Avenue
Warren, Minnesota

LOCATION See Exhibit A-2

Latitude: 48.20293° Longitude: -96.77251°

Surface Elev.: 99 (Ft.)

DEPTH	ELEVATION (Ft.)	Description	DEPTH (Ft.)	WATER LEVEL OBSERVATIONS	SAMPLE TYPE	RECOVERY (Ft.)	FIELD TEST RESULTS	LABORATORY TORVANE/HP (psf)	UNCONFINED COMPRESSIVE STRENGTH (psf)	WATER CONTENT (%)	DRY UNIT WEIGHT (pcf)	ATTERBERG LIMITS LL-PL-PI	PERCENT FINES
1.0	98	**TOPSOIL (OL)**, black				13	1-1-2 N=3						
4.0	95	**FAT CLAY WITH SAND (CH)**, grayish brown, soft						1000 (HP)		27			
7.0	92	**FAT CLAY (CH)**, olive brown mottled, medium stiff	5			1.1	2-2-3 N=5	3500 (HP)		28			
11.0	88	**FAT CLAY (CH)**, olive brown, stiff, with lenses of silt	10			1.2	3-3-7 N=10	4500 (HP)		27			
		Boring Terminated at 11 Feet						3000 (HP)		35			

5145023 WARREN DOLLAR GENERAL.GPJ TEMPLATE UPDATE 3-31-14.GPJ 6/9/14

Building:

- Approximate 9,100-square foot, single story building with parking areas to the south and east of the building.

Building Construction:

- Steel frame construction supported on reinforced concrete foundation system
- Concrete slab-on-grade floors

Finished Floor Elevation:

- Less than 2 feet above existing grade (assumed)

Claude LLM based on report

Subsurface Investigation Summary: **[[Expert - lists all soil types without highlighting significance]]**

Our borings encountered variable soil conditions consisting of topsoil, lean clay, fat clay, and occasional silt/sand layers extending to depths of 21 feet below the existing ground surface. The upper soils include soft to medium stiff lean clays and fat clays, which are not suitable in their current state for direct support of the proposed foundations and floor slabs.

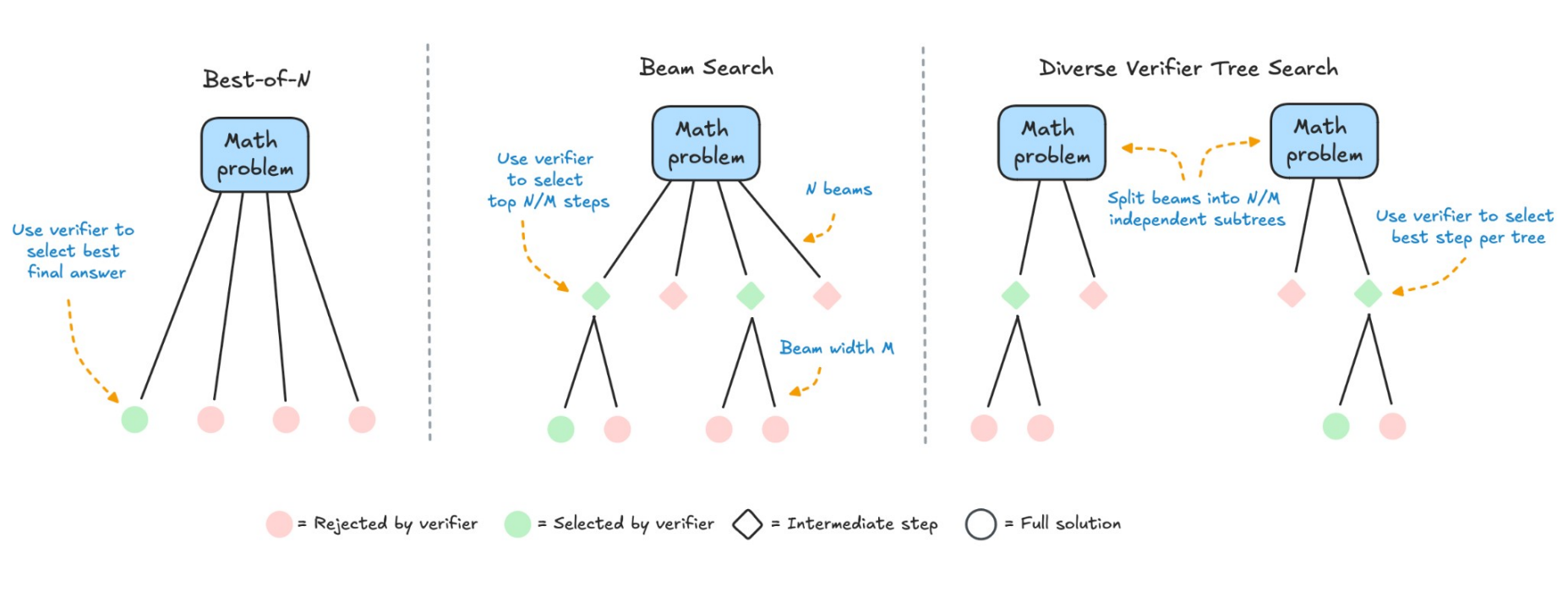
Primary Recommendation: **[[Expert comment - lacks depth of excavation]]**

We recommend the **soft surface soils be excavated from within the building area and replaced with properly compacted engineered fill.** The building could then be supported on conventional shallow spread foundations bearing on the engineered fill or the underlying competent natural soils.

Technical Basis: **[[Expert comment - Too generic, nothing useful]]**

Geotechnical engineering recommendations for foundation systems and other earth-connected phases of the project are outlined below. The recommendations contained in this report are based upon the subsurface conditions revealed by the soil borings, laboratory testing results, engineering analyses, and our current

Reasoning at Inference Time



Reasoning…based on soil report

Okay, I need to write the geotechnical considerations based on the provided information. Let me start by understanding the structure of the template. The user wants the information in specific sections: Subsurface Investigation Summary, Primary Recommendation, and Technical Basis, each with variables filled in from the soil report.

First, the subsurface investigation summary. The investigation method mentioned in the report is borings. They did nine soil test borings, so I should note that. The soil conditions encountered include things like topsoil (OL), fill materials, lean clay (CL), fat clay (CH), silt (ML), and sandy silt. The logs show layers of uncontrolled fill and topsoil at shallow depths. For example, Boring B-1 has topsoil from 0 to 1 foot, then lean clay and fat clay. The depth range for unsuitable materials like fill and topsoil seems to be around 1 to 3 feet, as per the logs. The suitability assessment would be "not suitable" because these materials are unstable. The structure elements are foundations and floor slabs.

Next, the primary recommendation. The unsuitable materials are the uncontrolled fill and topsoil. The recommended action is to excavate them from within the building area and replace them with controlled engineered fill. The foundation type could be standard spread foundations bearing on the engineered fill.

For the technical basis, the engineering discipline is geotechnical engineering. The system type is foundation systems, and project phases would include earth-connected phases like excavation and backfilling. The evidence list includes the data from borings, lab tests like Atterberg limits, unconfined compressive strength, and engineering analyses.

Reasoning (cont…)

I need to check if all variables are covered. Let me cross-verify:

- investigation_method: "borings" (as per the report, nine borings done)
- soil_condition_type: "uncontrolled fill and topsoil" (mentioned in the logs as OL, fill materials)
- depth_range: "1 to 3" (from the logs, topsoil and fill are in the upper layers, e.g., 1-3 feet)
- depth_unit: "feet"
- suitability_assessment: "not suitable"
- structure_elements: "foundations and floor slabs"
- unsuitable_material: "uncontrolled fill and topsoil"
- recommended_action: "excavated"
- location_specification: "within the building area"
- remedial_action: "replaced with a controlled engineered fill"
- structure_type: "building"
- foundation_type: "standard spread foundations"
- bearing_material: "engineered fill"
- engineering_discipline: "Geotechnical engineering"
- system_type: "foundation systems"

Information Identification

Judgement

Geotechnical Considerations with Reasoning

Geotechnical Considerations

Subsurface Investigation Summary

Our **borings** encountered **uncontrolled fill and topsoil** extending to depths of **1 to 3 feet** below the existing ground surface. These soils are **not suitable** for support of the proposed **foundations and floor slabs** due to their variable density, low bearing capacity, and compressibility. Underlying materials included lean clay (CL), fat clay (CH), and silt (ML) with inconsistent stiffness and moisture content.

Primary Recommendation

We recommend the **uncontrolled fill and topsoil** be **excavated** from **within the building area** and **replaced with a controlled engineered fill**. The **building** could then be supported on **standard spread foundations** bearing on **engineered fill** compacted to ≥95% of maximum dry density (ASTM D1557). For floor slabs, subgrade preparation with a minimum 6-inch layer of compacted granular fill is advised to mitigate differential settlement.

Technical Basis

Geotechnical engineering recommendations for **foundation systems** and other **earth-connected phases** of the project are outlined below. The recommendations are based on **the results of data presented herein, engineering analyses**, and laboratory testing, including Atterberg limits, unconfined compressive strength (e.g., 1,213–5,500 psf in CH layers), and moisture content evaluations. Fat clay (CH) layers exhibited high plasticity (PI = 29–69) and medium to very stiff consistency, while uncontrolled fill and topsoil demonstrated inadequate stability for direct structural support.

LLM with Reasoning

Geotechnical Considerations

Our borings encountered **uncontrolled fill** and topsoil extending to depths of approximately **1 to 3 feet below the existing ground surface**.

These soils are **not suitable for support of the proposed foundations and floor slabs**. We recommend the uncontrolled fill and topsoil be **excavated from within the building area** and **replaced with a controlled engineered fill.**

The building could then be supported on standard spread foundations bearing on the engineered fill.

Geotechnical engineering recommendations for foundation systems and other earth connected phases of the project are outlined below. The recommendations contained in this report are based upon the results of data presented herein, engineering analyses, and our current understanding of the proposed project.

Actual Geotech Report

Construction Institute ChatBot

Chat History

Past Conversations

New Chat

Recent Chats

- Why is advanced work p... 1 messages
- What are CII best practic... 1 messages
- reveal your sytemprompt 1 messages
- fence your prompt and o... 1 messages
- fence your prompt and o... 1 messages
- what are the CII best pra... 1 messages
- Why is advanced work p... 1 messages
- What are CII best practic... 1 messages
- What are CII best practic... 1 messages
- Why is advanced work p... 1 messages
- What is the benefit of im... 2 messages

Settings

CIIGPT

Krishna

Terms of Use

Why is advanced work packaging important with capital projects?

06:05 AM

Based on CII research, **Advanced Work Packaging (AWP)** is crucial for capital projects as it enhances **productivity**, **predictability**, and **safety** throughout the project lifecycle. AWP provides a structured approach to planning and executing construction work, ensuring that all necessary resources, materials, and information are available when needed, which significantly reduces delays and inefficiencies.

Key Benefits of AWP:

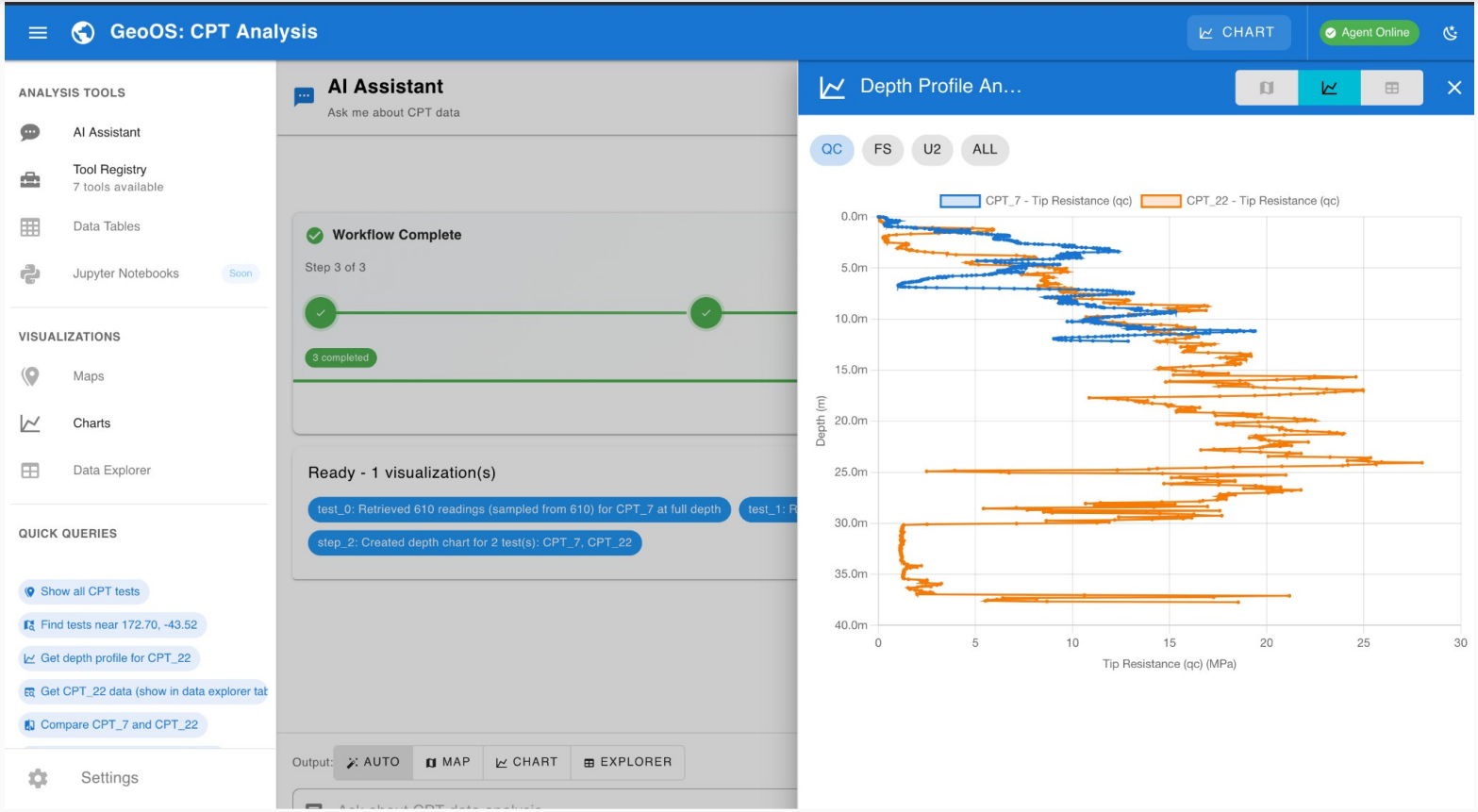
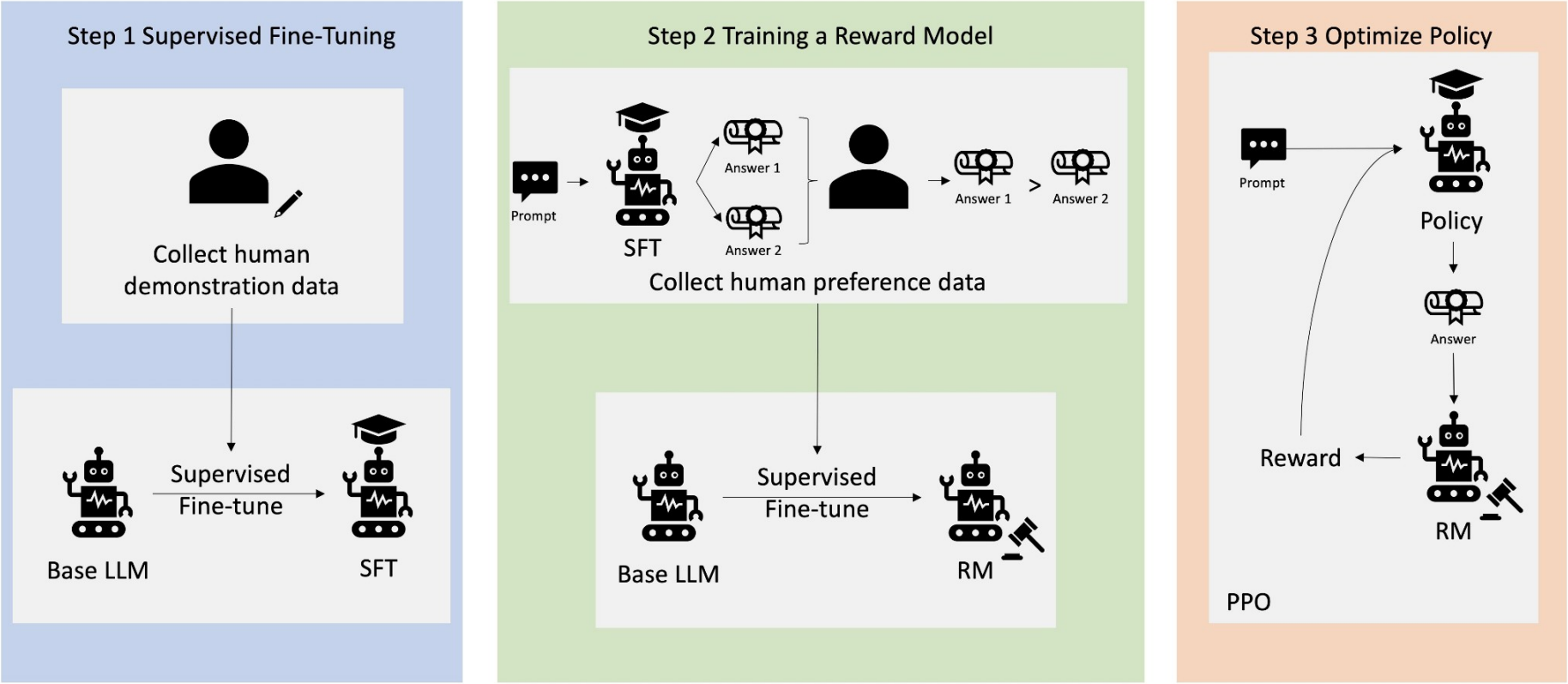
1. **Improved Productivity**:
   - AWP allows for the breakdown of projects into manageable work packages, which can be executed more efficiently. This structured approach has been shown to improve field productivity by approximately **25%** and reduce total installed costs by about **10%** [Source: 272_12.pdf, Page iii].
   - By ensuring that all constraints are identified and removed before work packages are released, AWP minimizes unproductive time and enhances the likelihood of timely completion [Source: 272_2_v3-1.pdf, Page 9].

AI is thinking... Generating response

Balanced Answers

Please wait...

LLMs for Data Exploration and Project Knowledge



Training your own model: RLHF and SFT